



Translating objects

Task A: Translating a single point from a description

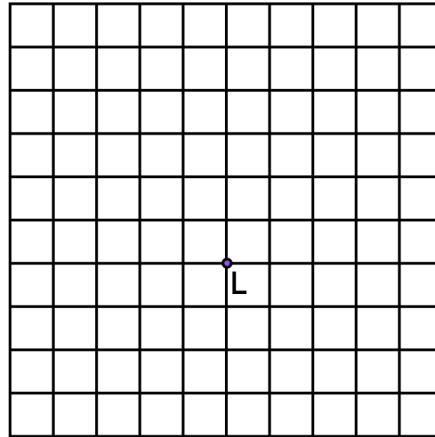
- 1) Mark the image of point L after the following translations:

a) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

b) $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$

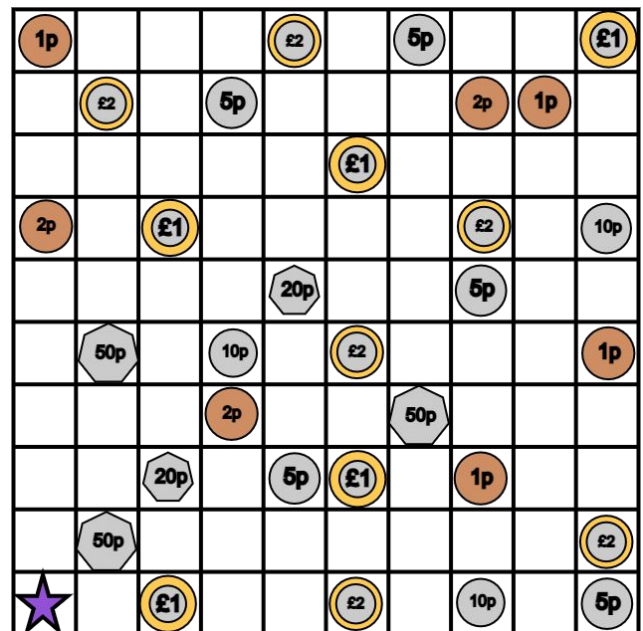
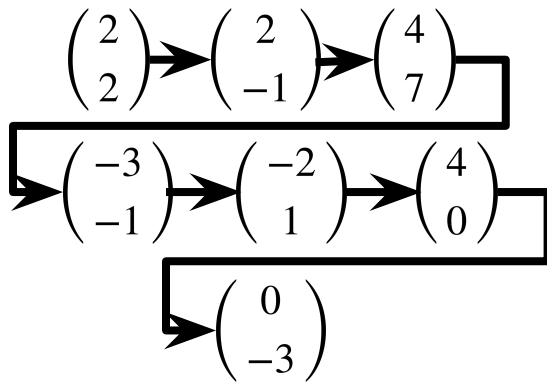
c) $\begin{pmatrix} -1 \\ -3 \end{pmatrix}$

d) $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$



- 2) Work out the total money collected when following this vector trail.

Start on the star.



- 3) Investigate the total money you can collect using all of the vectors but in any order.

Start on the star.

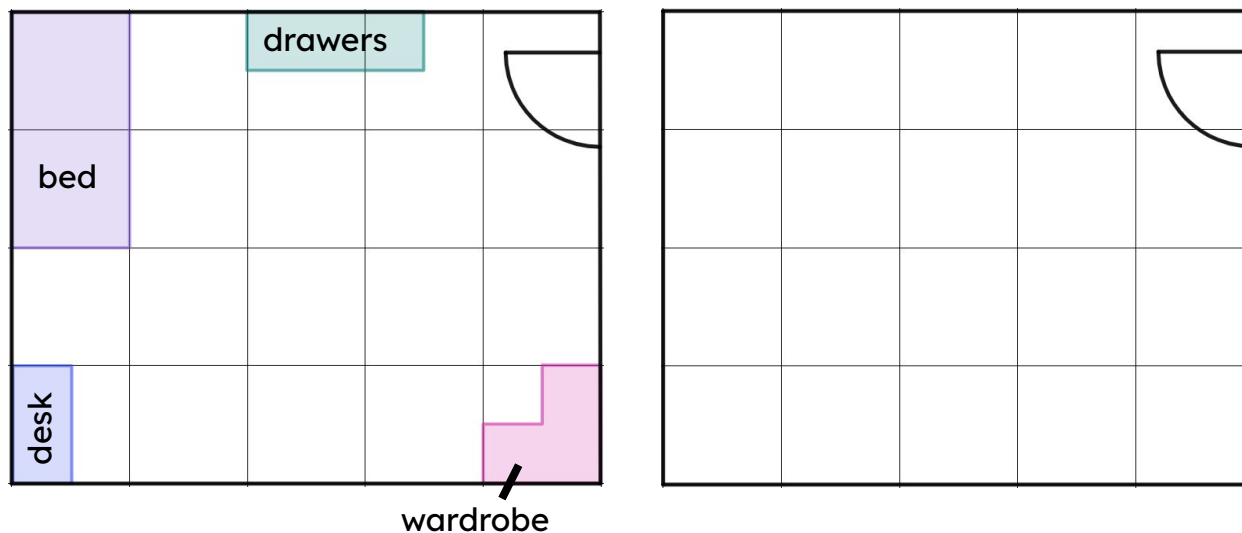
$A = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$
 $B = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$
 $C = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$
 $D = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$
 $E = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$

$F = \begin{pmatrix} -3 \\ -1 \end{pmatrix}$
 $G = \begin{pmatrix} 0 \\ -3 \end{pmatrix}$



Task B: Translating more complex objects

- 1) Here is a plan view of a bedroom. By only translating the objects, redesign the layout.



- 2) Complete the translations, always start with triangle A.

a) $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ b) $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$

c) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ d) $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$

e) $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ f) $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$

